

Day 2

Chapter 2: First Order Differential Equations

2.2: Separable Equations

Definition: Separable Function

A function $z = f(x, y)$ is called separable if we can write f as a product of two functions, where one is a function of x and the other is the function of y , that is, $z = f(x, y) = g(x)h(y)$.

Consider the first order differential equation:

$$\frac{dy}{dx} = f(x, y) \dots \dots (1)$$

The DE (1) is called separable if the right-hand side function is separable, that is,

$$\frac{dy}{dx} = g(x)h(y) \dots \dots (2)$$

We can solve the differential equation (2) by separating variables.

From DE (2):

$$\begin{aligned} \frac{dy}{dx} &= g(x)h(y) \Rightarrow dy = g(x)h(y) dx \\ &\Rightarrow \frac{1}{h(y)} dy = g(x)dx \end{aligned}$$

Now, integrating both sides with respect to respective variables, we get

$$\int \frac{1}{h(y)} dy = \int g(x)dx .$$

Examples

Example:1

Solve the DE:

$$e^x y \frac{dy}{dx} = e^{-y} + e^{-2x-y} .$$

Solution: Consider the DE:

$$e^x y \frac{dy}{dx} = e^{-y} + e^{-2x-y}$$

$$\Rightarrow e^x y \frac{dy}{dx} = e^{-y}(1 + e^{-2x})$$

$$\Rightarrow e^x y dy = e^{-y}(1 + e^{-2x}) dx ; \text{ divide both sides by } e^x e^{-y}$$

$$\Rightarrow \frac{e^x y}{e^x e^{-y}} dy = \frac{e^{-y}(1 + e^{-2x})}{e^x e^{-y}} dx$$

$$\Rightarrow ye^y dy = \left(\frac{1}{e^x} + \frac{e^{-2x}}{e^x} \right) dx$$

$$\Rightarrow ye^y dy = (e^{-x} + e^{-3x}) dx ; \text{ Integrating both sides with respect to respective variables}$$

$$\Rightarrow \int ye^y dy = \int (e^{-x} + e^{-3x}) dx$$

$$\Rightarrow ye^y - e^y = -e^{-x} - \frac{1}{3}e^{-3x} + C$$

$$\therefore ye^y - e^y + e^{-x} + \frac{1}{3}e^{-3x} = C .$$

Example:2

Solve the DE:

$$\frac{dy}{dx} = \frac{xy + 2y - x - 2}{xy - 3y + x - 3} .$$

Solution: Consider the DE:

$$\frac{dy}{dx} = \frac{xy + 2y - x - 2}{xy - 3y + x - 3}$$

$$\Rightarrow \frac{dy}{dx} = \frac{(x+2)(y-1)}{(x-3)(y+1)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{(x+2)}{(x-3)} \cdot \frac{(y-1)}{(y+1)}$$

$$\Rightarrow dy = \frac{(x+2)}{(x-3)} \cdot \frac{(y-1)}{(y+1)} dx$$

$$\Rightarrow \frac{(y+1)}{(y-1)} dy = \frac{(x+2)}{(x-3)} dx$$

$$\Rightarrow \int \frac{(y+1)}{(y-1)} dy = \int \frac{(x+2)}{(x-3)} dx$$

$$\Rightarrow \int \frac{y-1+2}{y-1} dy = \int \frac{x-3+5}{x-3} dx$$

$$\Rightarrow \int \left[1 + \frac{2}{y-1} \right] dy = \int \left[1 + \frac{5}{x-3} \right] dx$$

$$\Rightarrow y + 2 \ln|y-1| = x + 5 \ln|x-3| + C_0$$

$$\Rightarrow \ln e^y + 2 \ln|y-1| = \ln e^x + 5 \ln|x-3| + \ln e^{C_0}$$

$$\Rightarrow 2 \ln|y-1| - 5 \ln|x-3| = \ln(e^{C_0} \cdot e^x) - \ln e^y$$

$$\Rightarrow \ln(y-1)^2 - \ln(x-3)^5 = \ln(c \cdot e^x) - \ln e^y \quad ; \quad \text{for } x > 3 \text{ and } C = e^{C_0}$$

$$\Rightarrow \ln \left(\frac{(y-1)^2}{(x-3)^5} \right) = \ln \left(\frac{c \cdot e^x}{e^y} \right)$$

$$\therefore \frac{(y-1)^2}{(x-3)^5} = c \cdot e^{x-y} \quad ; \quad \text{for } x > 3.$$

Example:3

Solve the IVP:

$$\frac{dy}{dx} = e^{-x^2}, \quad y(3) = 5.$$

Solution: Given IC $y(3) = 5$, that is, the initial value of x is $x = 3$. The function $g(x) = e^{-x^2}$ is continuous on the interval $(-\infty, \infty)$, but its antiderivative is NOT an elementary function. So, consider an arbitrary interval $I = [\text{initial value, arbitrary } x] = [3, x]$ and using the dummy variable t in the DE, we get:

$$\frac{dy}{dt} = e^{-t^2} \dots (1)$$

Now, integrating both sides of (1) with respect to t over the interval $I = [3, x]$:

$$\int_3^x \frac{dy}{dt} dt = \int_3^x e^{-t^2} dt$$

$$\Rightarrow \int_3^x dy = \int_3^x e^{-t^2} dt$$

$$\Rightarrow [y(t)]_3^x = \int_3^x e^{-t^2} dt$$

$$\Rightarrow y(x) - y(3) = \int_3^x e^{-t^2} dt$$

$$\Rightarrow y(x) - 5 = \int_3^x e^{-t^2} dt \quad ; \quad y(3) = 5$$

$$\therefore y(x) = 5 + \int_3^x e^{-t^2} dt .$$

Example:4

Solve the IVP:

$$\frac{dy}{dx} = e^{5x^2} , \quad y(x_0) = y_0 .$$

Example:4

Solve the IVP:

$$\frac{dy}{dx} = ye^{-x^2} , \quad y(4) = 1 .$$

Exercise: 2.2 Page-51 (separable Variable)

Exercises number: 11- 15,18,20, 23-28, 29,30.